

Heaven's light is our guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Mechatronics Engineering

Lab Report

Experiment no.:06

***Experiment name: Designing and Solving Real life problems using PLC
programming***

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MTE 4102

Automation Sessional

Experiment no.:06

Experiment name: Designing and Solving Real life problems using PLC programming

Objectives:

- To develop problem-solving skills by translating real-world industrial processes into PLC-based control logic.
- To understand the use of timers, counters, sensors, and actuators in automation systems.
- To design PLC programs for fire alarm systems, machine cooling, conveyor-based bottle filling, gate control, tank mixing, overheating alarms, and traffic lights.
- To analyze system responses and ensure reliability, safety, and efficiency in simulated industrial processes.

Theory:

A Programmable Logic Controller (PLC) is an industrial digital computer used to automate electromechanical processes such as factory assembly lines, machines, and building systems. PLCs are designed for multiple input and output arrangements, extended temperature ranges, immunity to electrical noise, and resistance to vibration and impact.

PLC programming is typically done using Ladder Logic, Function Block Diagrams, or Structured Text. The logic involves conditions, timers, counters, and sequencing to perform tasks automatically. For example:

In a fire alarm system, input sensors detect smoke/fire, and outputs such as buzzers and lights are activated.

In a bottle-filling system, timers synchronize conveyor stoppage and filling operations.

Traffic light control uses sequential logic with timers to simulate real-world traffic flow.

Through this lab, each problem reflects a practical application of PLCs in industries such as manufacturing, safety systems, and transportation.

Required Apparatus:

1. Laptop
2. OpenPLC Editor (PC Software)

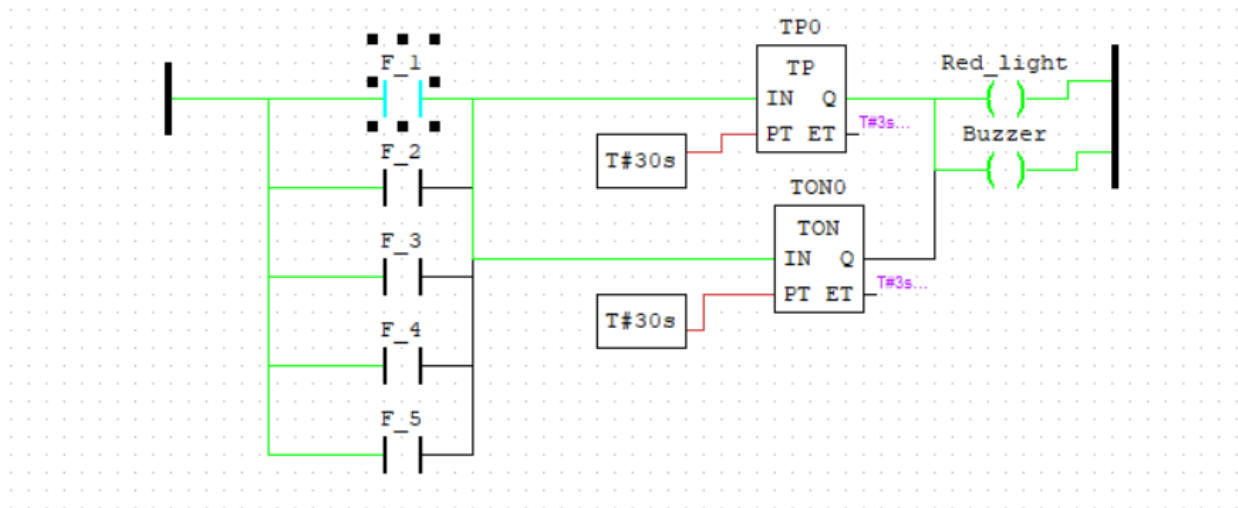
Problem Statements & Ladder Diagram:

Problem 1: 5-Floor Building Fire Alarm System

5-Floor Building Fire Alarm System

- Each floor has one fire sensor.
- If any one sensor detects fire:
 - It triggers a buzzer and a red light.
- The alarm will remain active for 30 seconds once triggered.
- If the fire is no longer detected during the 30 seconds, the alarm will turn off immediately.

Ladder Diagram:

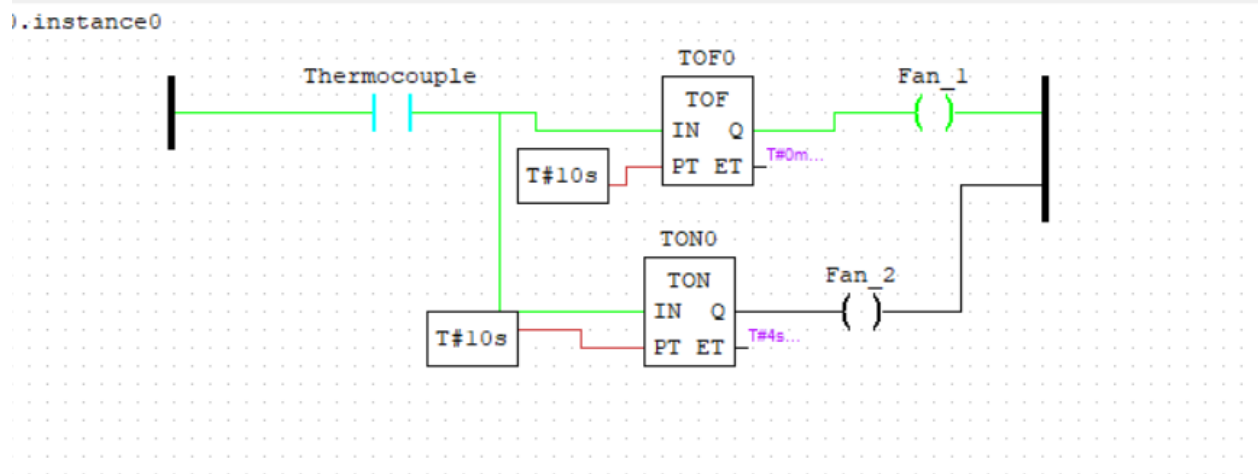


Problem 2: Machine Cooling System

Machine Cooling System

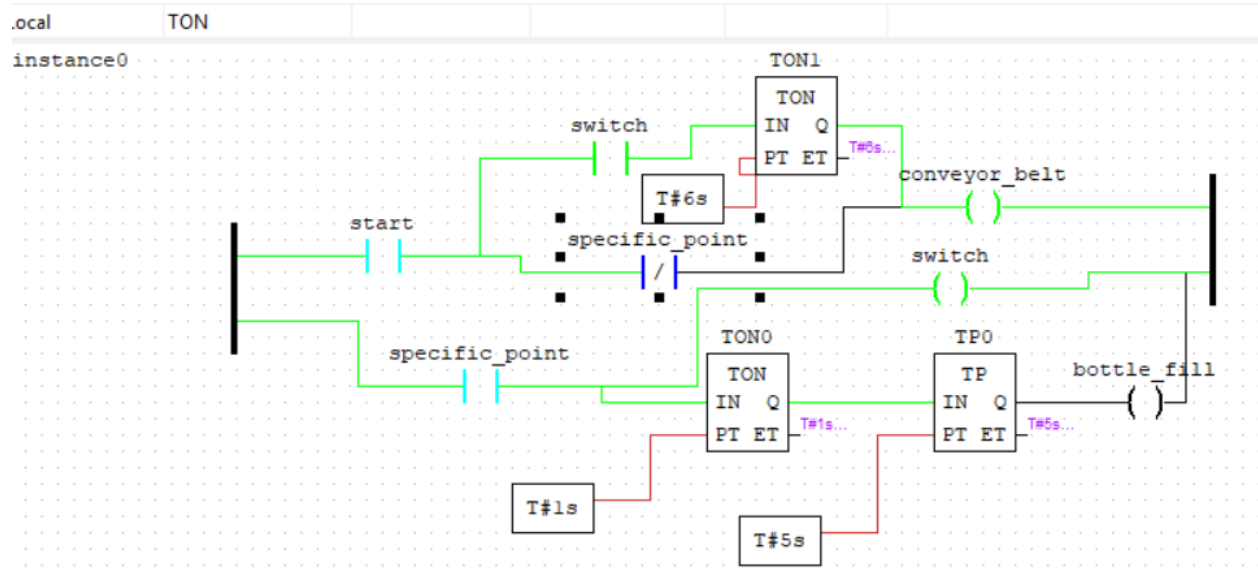
- The machine has two fans for cooling purposes.
- If the machine's temperature exceeds the threshold:
 - Fan 1 will start.
 - After 10 seconds of Fan 1 activation, Fan 2 will start.
- If the temperature drops below the threshold:
 - Fan 2 will stop immediately.
 - Fan 1 will stop 10 seconds later.

Ladder Diagram:



Problem 3: Conveyor Belt Bottle Filling

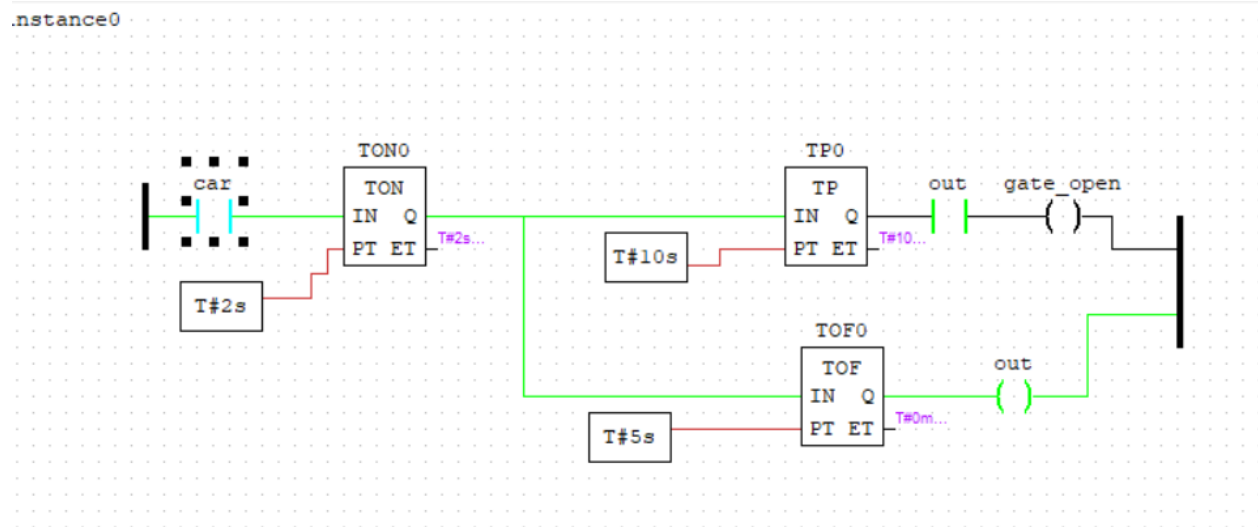
- When an empty bottle reaches a specific point, the conveyor belt will stop.
- After 1 second, liquid will be filled into the bottle for 5 seconds.
- Then, after another 1 second, the conveyor belt will start again, allowing the next bottle to come.



Problem 4: Automatic Gate System

- If a car arrives, the gate will open after 2 seconds.
- The gate will remain open for 10 seconds.
- During these 10 seconds, if no car is detected, the gate will close after an additional 5 seconds.

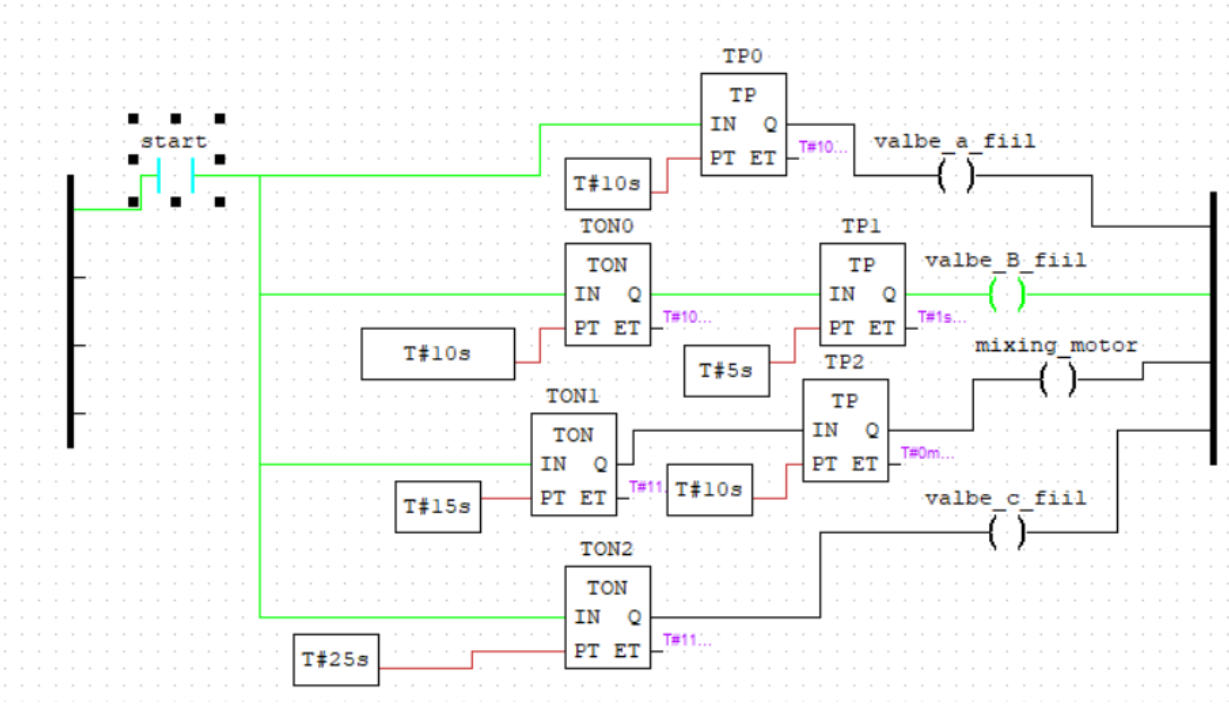
Ladder Diagram:



Problem 5: Tank Mixing System (Liquid 1 and 2)

- When the Start switch is pressed:
 - Valve A will open, and Liquid 1 will enter the tank for 10 seconds.
 - Then, Valve A will close, and Valve B will open.
 - Liquid 2 will enter the tank for 5 seconds.
 - Valve B will then close.
 - A mixing motor will turn on for 10 seconds to mix the liquids.
 - After mixing, Valve C will open, and the mixed liquid will exit the tank.

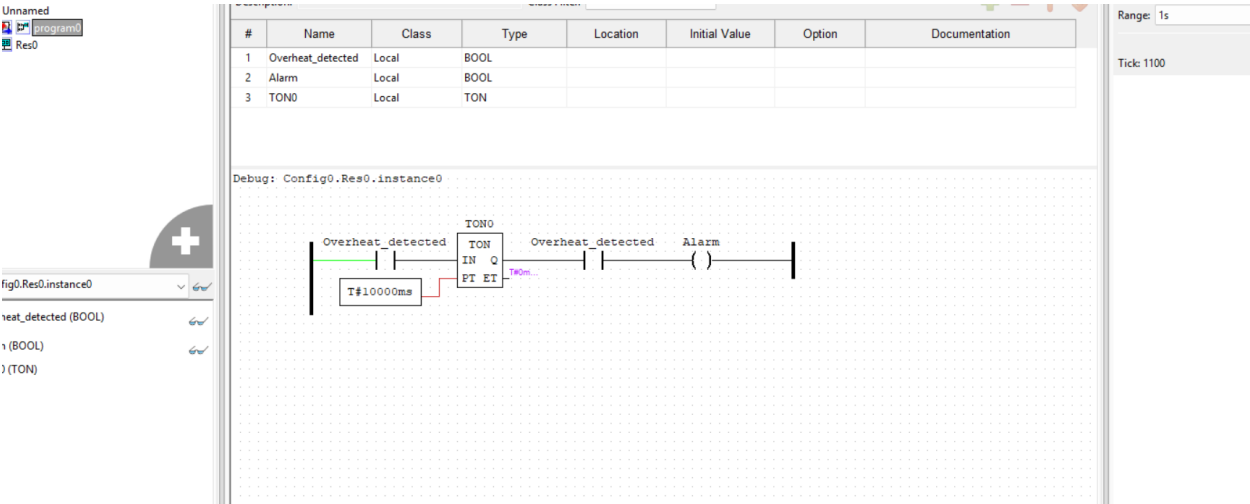
Ladder Diagram:



Problem 6: Machine Overheat Alarm

- If the machine overheats, and the overheat condition persists for 10 seconds,
 - o An alarm will be triggered.

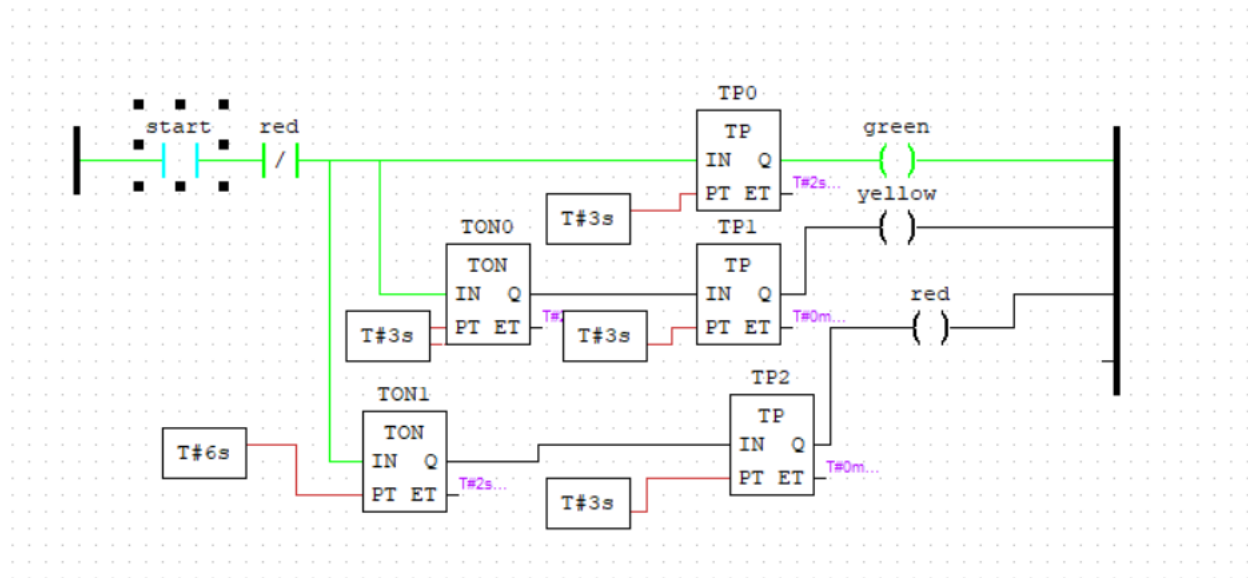
Ladder Diagram:



Problem 7: Traffic Light Control System

- When the system is turned on:
 - o Green light will turn on for 10 seconds.
 - o Then, Yellow light will turn on for 3 seconds.
 - o Then, Red light will turn on for 10 seconds.
- This cycle will continue in sequence as long as the system is on.
- If the system is turned off, the lights will stop.

Ladder Diagram:

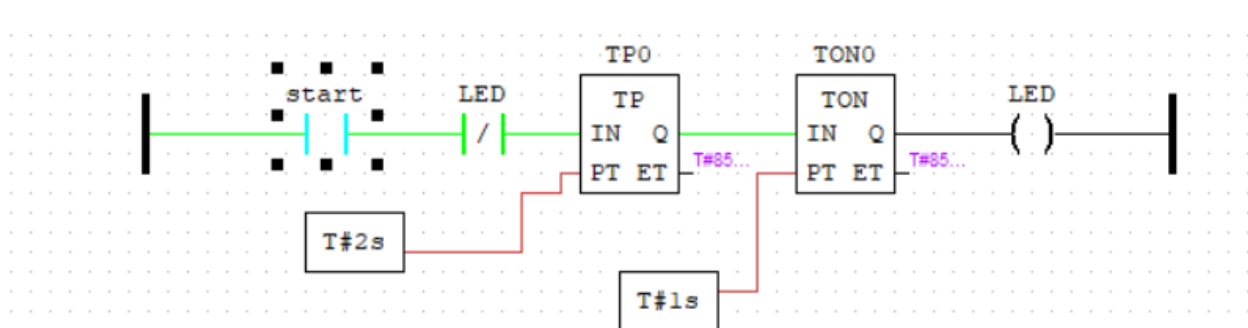


Problem 8: Blinking Light System

stem

- The light will blink by turning on for 1 second and off for 1 second, repeatedly.
- If the Stop button is pressed, the light will turn off and stop blinking.

Ladder Diagram:



Discussion:

The lab allowed us to implement and simulate different automation problems that closely resemble real-world industrial applications. Each problem highlighted how PLCs are used for reliability and precision. For instance, the fire alarm system demonstrated how safety-critical systems can be automated to respond quickly to hazards. The machine cooling system emphasized the importance of sequential timing in protecting machinery from overheating. The conveyor belt bottle filling problem showed how synchronization between sensors, actuators, and timers ensures efficiency in production lines.

Similarly, the automatic gate system highlighted real-time response and conditional operation based on car detection. The tank mixing system combined sequential valve operation, mixing, and output, reflecting chemical and food industries' automation needs. The overheating alarm emphasized delay-based monitoring, preventing false alarms from temporary fluctuations. Lastly, the traffic light system demonstrated continuous sequential control, a common example of PLC application in transportation.

By solving these problems, we gained a practical understanding of how PLC logic is structured to ensure smooth and safe operation. The use of timers, sequential control, and sensor-based logic was common across most problems, showcasing their importance in industrial automation.

Conclusion:

This lab successfully demonstrated how PLC programming can be used to design and solve real-life automation problems. By working through seven different problems, we applied logical thinking, timer-based sequencing, and conditional control to mimic industrial processes. The exercises highlighted the versatility of PLCs in safety, manufacturing, and transportation applications. Overall, the lab enhanced our understanding of automation, improved our problem-solving skills, and prepared us to handle practical industrial challenges using PLCs.